DIGITAL DASHBOARD

User Manual

Unified CUSUM and geospatial analysis workflow

Documentation revision	June 2026
Language	English
Status	Aligned with the supplied current application source

Purpose. This document describes the current workflow and implementation represented by the supplied project files. It replaces older instructions that referred to previous navigation, storage, or output structures.

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1. What the Dashboard does

The Digital Dashboard combines the CUSUM RAP and Geospatial RAP workflows in one interface. One upload and one set of shared filters feed early-warning plots, prevalence maps, LISA maps, tables, prepared datasets, and downloads.

Navigation page	Purpose
Home	Overview and recommended workflow.
Dashboard	Upload, filters, CUSUM parameters, Run Analysis, outputs, tables, and prepared data.
Display	Choose visible plots/maps, layout, table location, font size, decimals, and optional reactive updates.
Help	Workflow summary and documentation links.

Two different location behaviors. The Locations selector limits CUSUM units. The current Geospatial workflow still maps all units available at the selected geographic level and time window.

2. Input files and file management

Use prepared API-POP-compatible delimited files. The file picker lists several extensions, but the supplied importer reads CSV-formatted text and detects comma or semicolon separators. CSV is therefore the reliable supported format for this build.

Column	Requirement	Meaning
country	Yes for current validation	Country value. Current source fixes active_country to Argentina.
year	Yes	Epidemiological year, 2000-2100.
week	Yes	Epidemiological week, 1-53.
level1..level4	At least one	Geographic hierarchy columns used by the configured level.
n_cases	Yes	Observed cases; aliases cases, case_count, count, n.
population	Maps and manual mu0	Denominator for maps and manual rate conversion; aliases n_population, pop, denominator.

Each uploaded file is stored in the current session with its row count, column count, epidemiological range, modification time, and size. New files are active by default.

- Manage files: activate or deactivate files without re-uploading.
- Remove inactive files: deletes unchecked files from the current session.
- Clear all files: removes all uploaded data and analysis results from the session.
- Changing active files invalidates existing CUSUM and geospatial results; run the analysis again.

No database write. The current Dashboard analyses uploaded session data. It does not append the upload to a persistent epidemiological database.

3. Shared filters and CUSUM settings

Control	Effect

Geographic Level	Sets the administrative level used by both analytical engines.
Detection Period	Number of recent weeks used by CUSUM; UI range 8-260, default 52.
Locations	Selected by default for all available units. Changes affect CUSUM only in the current implementation.
Year	Year used for geospatial maps and the available week slider.
Map Time Window	Single week or inclusive week range. Range mode enforces at least two weeks when possible.
Expected Frequency	Automatic expected count by unit, or manual weekly rate per 100,000 converted using each unit's population.
Target ARL0	Used to display a recommended h.
Relative Risk	Used to calculate k.
h Threshold	CUSUM alarm threshold.
Run Analysis	Refreshes both CUSUM and Geospatial outputs using the current active data and selections.

Calculated k and recommended h appear under the CUSUM settings when enough information is available.

> **Run required after structural changes.** Changes to active data, Geographic Level, or Locations require Run Analysis. Display settings can be changed before or after a run.

4. Configure the Display page

Display settings may be configured before the first analysis or changed afterward. They control presentation, not the underlying data.

Setting	Options and behavior
Visible outputs	CUSUM overview, CUSUM series, CUSUM process, Prevalence map, LISA map.
Output layout	One or two columns.
Results table location	Inside Output, or in a separate Output Table tab within Dashboard.
Display font size	Small, Medium, Large, or Extra large.
Decimal places	0-6 decimals in on-screen tables; CSV precision is unchanged.
Reactive display	After a successful run, CUSUM parameter changes and map year/week/time-mode changes can refresh results automatically.

Reactive display does not automatically apply active-file, geographic-level, or location changes. Those changes still require Run Analysis.

5. Run and review the Dashboard

1. Upload and activate the required CSV files.
2. Set Geographic Level, Locations, Detection Period, Year, and Map Time Window.
3. Configure CUSUM parameters.
4. Optionally configure Display.
5. Select Run Analysis.
6. Review Output, Prepared Data, and the optional Output Table tab.

CUSUM outputs

- CUSUM overview: alarm heatmap across selected CUSUM units.
- CUSUM series plot: observed and expected values for the selected detailed unit.
- CUSUM process plot: cumulative statistic and threshold.
- CUSUM weekly results and unit-level reference tables.

Geospatial outputs

- Prevalence map for all matched units at the selected level.
- LISA cluster map using the same spatial units and time window.
- Global Moran summary where the data support it.
- Geospatial results table.

Prepared Data

The Dashboard exposes two prepared datasets: CUSUM prepared data and Active standardized data. Each can be downloaded as CSV. The tables support a selected-column Search by filter.

Download availability. Download buttons appear only after the corresponding result exists. Map image export also depends on optional screenshot packages in the deployment.

6. How the two analyses use time and geography

Dimension	CUSUM	Geospatial
Locations	Restricted to the selected Locations.	Current implementation ignores the Locations subset and uses all units at Geographic Level.
Detection period	Controls the recent CUSUM window.	Not used.
Year	CUSUM preparation uses the available weekly history rather than the map Year selector.	Filters the map data to the selected year.
Week selection	Not controlled by Map Time Window.	Uses the selected single week or inclusive range.
Population	Manual: required; auto: optional.	Required for prevalence.
Geographic level	Builds CUSUM units at the selected level.	Loads and joins the configured layer for the selected level.

Avoid misinterpretation. Do not assume a selected Locations subset also clips the maps. The current maps remain a complete level-wide comparison.

7. Interpretation and exports

- CUSUM alarms are early warnings requiring epidemiological review.
- Prevalence maps are descriptive and depend on numerator and denominator quality.
- Global Moran describes overall spatial autocorrelation; LISA describes local clusters/outliers.
- Display choices do not change the analytical values.

Export	Typical content
CUSUM PNG	Selected heatmap, series, or process plot.
Geospatial map PNG	Current map when screenshot support is installed.

CUSUM CSV	Weekly results or unit reference values.
Geospatial CSV	Joined spatial-analysis results.
Prepared CUSUM CSV	Dataset used by CUSUM.
Active Standardized CSV	Combined active data after column normalisation and code padding.

8. Troubleshooting

Problem	What to check
Uploaded XLSX/TSV does not load	Convert to comma- or semicolon-delimited CSV; the current reader is CSV based.
CUSUM completes but maps fail	Population, GeoPackage, map IDs, selected level, or year/week coverage may be missing.
Maps show all areas despite selected Locations	Expected current behavior; Locations restrict CUSUM only.
Range snaps to two weeks	Range mode enforces at least a two-week span when more than one week exists.
Outputs look stale	Run Analysis after active-data, level, or location changes.
Map download missing	Optional screenshot packages may not be installed.
Documentation link fails on Linux	Use the exact case-sensitive filenames included in the updated docs package.

Escalation package. Provide active filenames, filter values, CUSUM parameters, Display settings, exact error text, and whether CUSUM, Geospatial, or both failed.

9. Glossary

Term	Meaning
Active file	An uploaded file currently included in the combined session data.
Standardized data	Uploaded columns after alias normalization, numeric conversion, trimming, and code padding.
Reactive display	Automatic rerun after eligible parameter changes, once an initial result exists.
Output Table	Optional Dashboard tab used when table placement is set to separate.
Prepared CUSUM data	Aggregated and completed weekly dataset used by the CUSUM engine.
Map Time Window	Year-specific week or week range used only by Geospatial analysis.